CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in turning business problems into working AI solutions, from LLM workflows and prompt engineering to API and MCP integrations and quick proofs-of-concept. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by consulting work with business stakeholders and hands-on Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an LLM-powered knowledge-extraction workflow end-to-end, from data ingestion and chunking to prompt engineering and indexing, reporting directly to the CTO and working closely with SAP domain experts on which source data feeds the knowledge graph.
- Developed Expert Interface, a web app (React frontend, Python microservices backend) on Azure that lets non-developer domain experts validate and approve knowledge-graph changes themselves.
- Built multi-agent LLM workflows (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Chose a deterministic, code-based pipeline over LLM calls for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) service exposing the knowledge graph to AI agents with millisecond-latency retrieval, integrated into Claude Code for agentic workflows.
- Used Claude Code, custom skills and agentic coding workflows to ship working first versions fast.
- **Tech stack: Python, LLMs (LiteLLM), Prompt Engineering, MCP, LangGraph, LangChain, RAG, GraphRAG, FastAPI, React, Azure, Docker, Kubernetes, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Sat with business stakeholders to understand their decision-making workflows, advised on how agentic AI could support them, and translated their requirements into a concrete solution approach.
- Designed and deployed an agentic AI application with LangGraph that routes each request to vector database search, general LLM inference or summarization.
- Developed a React front-end and a Python/FastAPI REST backend, tested the APIs with Postman, containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, REST APIs, Postman, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated generation and retrieval evaluation for the PrismaChat RAG assistant with RAGAS and PytrecEval, comparing LLM answers to chemist-provided ground truth.
- Built AWS infrastructure with Lambda, SQS and DynamoDB via CDK to run the evaluations automatically.
- Benchmarked multilingual embedding models on over 1M data chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, retrieval fine-tuning).

- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data.
- Delivered an interactive Streamlit dashboard so chemists could simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Ops MCP Server | MCP | Claude | Slack | HubSpot API | Notion | TypeScript

- MCP server that gives Claude and a Slack bot typed, permission-checked tools over HubSpot deals and contacts and retrieval over Notion process docs, so a business team's everyday questions get answered from live data instead of manual lookups.
- Slack webhooks with request signing and OAuth, Zod schemas for every tool, role-based access per team, prompts and settings the team can adjust without code, a handover guide for running it, a small eval set in CI with GitHub Actions, packaged as a Claude Code plugin.

TECHNICAL SKILLSET

- **LLMs & Automation:** LLMs (Claude, LiteLLM), Prompt Engineering, MCP, LangGraph Agentic Workflows, LangChain, RAG, GraphRAG, Claude Code
- **Programming & Integrations:** Python, SQL, APIs (REST, HTTP/JSON), FastAPI, Postman, Python Microservices, React (beginner)
- **Data:** Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search), Pandas, Streamlit
- **Cloud & Delivery:** Azure, AWS (S3, EC2, Lambda, SQS, DynamoDB, CDK), Docker, Kubernetes, Git, GitLab CI/CD
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, Hugging Face Transformers, RAGAS
- **Methodologies:** Agile, stakeholder requirements analysis, requirements to solution approach

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)